SECTION 1 (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme**:

Full Marks : $+3$ If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.1 Let $\vec{a}$, $\vec{b}$ be two vectors, and let P, Q and R be the points with position vectors $\vec{a}$, $\vec{b}$ and $\vec{a}+\vec{b}$, respectively, with respect to the origin O. If $|\vec{a}+\vec{b}| = \sqrt{21}$, $|\vec{a}-\vec{b}| = 3$, and $\vec{a}$ and $(\vec{a}-\vec{b})$ are perpendicular to each other, then the area of the triangle OPR is

(A)	$\sqrt{3}$	(B)	$\frac{\sqrt{3}}{2}$	(C)	$\frac{3\sqrt{3}}{2}$	(D)	$\frac{3}{2}$

Q.2 Let T be the tangent to the parabola $y^2 = 16x$ at the point $(64, 32)$. Let L be the tangent to the same parabola at another point (x_1, y_1) on the parabola. If L and T are perpendicular to each other, then the distance between the point (x_1, y_1) and the focus of the parabola, is

(A)	$\frac{15}{4}$	(B)	4	(C)	$\frac{17}{4}$	(D)	5

Q.3 Let $y : (-\infty, \infty) \to (0, \infty)$ be the solution of the differential equation

$$\frac{dy}{dx} = \frac{e^{5x}y^3 + y^3}{e^x + e^x y^4},$$

satisfying $y(0) = \frac{1}{\sqrt{2}}$. Then the value of $y(\log_e 2)$ is

(A)	$\sqrt{\frac{5+\sqrt{35}}{2}}$	(B)	$\sqrt{\frac{7+\sqrt{53}}{2}}$
(C)	$\frac{7+\sqrt{53}}{2}$	(D)	$\frac{5+\sqrt{35}}{2}$

Q.4 The value of the definite integral

$$\int_0^2 \frac{1}{3^x + 3} dx$$

is

(A)	$\frac{1}{2}$	(B)	$\frac{1}{3}$	(C)	$\frac{\log_e 3}{3}$	(D)	$\frac{\log_e 3}{2}$

SECTION 2 (Maximum Marks: 20)

- This section contains **FIVE (05)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated **according to the following marking scheme**:

  Full Marks : $+4$ **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : $+3$ If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : $+2$ If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : $+1$ If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
  choosing ONLY (A), (B) and (D) will get $+4$ marks;
  choosing ONLY (A) and (B) will get $+2$ marks;
  choosing ONLY (A) and (D) will get $+2$ marks;
  choosing ONLY (B) and (D) will get $+2$ marks;
  choosing ONLY (A) will get +1 mark;
  choosing ONLY (B) will get +1 mark;
  choosing ONLY (D) will get +1 mark;
  choosing no option (i.e. the question is unanswered) will get 0 marks; and
  choosing any other combination of options will get -1 marks.

Q.5

Let $\mathbb{R}$ denote the set of all real numbers. Consider the polynomial function $f: \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \frac{d^{10}}{dx^{10}}((x^2-1)^{10}), \qquad \text{for all } x \in \mathbb{R}.$$

Here $\frac{d^{10}}{dx^{10}}((x^2-1)^{10})$ is the 10th order derivative of the function $(x^2-1)^{10}$.

Then which of the following statements is (are) TRUE ?

(A)	The coefficient of x^8 in the polynomial $f(x)$ is $(-10)\left(\frac{18!}{8!}\right)$
(B)	The value of $f(1)+f(-1)$ is equal to $10!\,2^{11}$
(C)	The degree of the polynomial $f(x)$ is 10
(D)	The constant term of the polynomial $f(x)$ is $-\left(\frac{10!}{5!}\right)$

Q.6 Let $a,\ b,\ c$ be positive integers in arithmetic progression such that the equation

$$ax^2 + bx + c = 0$$

has only integer solutions.

Then which of the following statements is (are) TRUE ?

(A)	$c - b$ is an integer multiple of a
(B)	Both the roots of the equation $ax^2 + bx + c = 0$ are odd integers
(C)	If $c = 15$, then $ab = 8$
(D)	If $b = 8$, then $x = 3$ is a root of the equation $ax^2 + bx + c = 0$

Q.7 Let L be the straight line joining the points $P(1, 2, -1)$ and $Q(2, 3, 1)$. Let S be the foot of the perpendicular drawn from the point $R(4, -1, 5)$ to the line L. Another line passing through R intersects L at a point T such that the point S divides the line segment PT internally in the ratio $|PS| : |ST| = 1 : 2$, where $|PS|$ and $|ST|$ are the lengths of the line segments PS and ST, respectively.

Then which of the following statements is (are) TRUE ?

(A)	The orthocentre of the triangle PRT is $\left(\frac{23}{5}, -4, \frac{31}{5}\right)$
(B)	The orthocentre of the triangle PRT is $(4, 3, 5)$
(C)	The area of the triangle PRT is $6\sqrt{5}$
(D)	The area of the triangle PRT is $18\sqrt{5}$

Q.8 Let $y = f(x)$ be the real valued function defined on the interval $(0, \infty)$, satisfying $y(1) = 0$ and the differential equation

$$x\frac{dy}{dx} = y - x^3 .$$

Then which of the following statements is (are) TRUE ?

(A)	The function f has a local minimum at $x = \frac{1}{\sqrt{3}}$
(B)	The function f has a local maximum at $x = \frac{1}{\sqrt{3}}$
(C)	The function f is increasing in the interval $(1,\ 2)$
(D)	If $g(x) = 4x^3 - 5x^2 + \frac{3}{2}x$ for $x > 0$, then the number of elements in the set $\{x \in (0, \infty) :\ f(x) = g(x)\}$ is 2

Q.9

Let $\mathbb{R}$ denote the set of all real numbers and let $i = \sqrt{-1}$. Consider the matrices

$$S = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ and } T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Let a, b, c, d be real numbers such that

$$ST = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Let

$$H = \{ x + iy : \ x, y \in \mathbb{R} \text{ and } y > 0 \}.$$

Then which of the following statements is (are) TRUE ?

(A)	$\frac{b+ia}{d+ic} = i$
(B)	If $\omega = \frac{-1+i\sqrt{3}}{2}$, then $\frac{a\omega+b}{c\omega+d} = \omega$
(C)	If m is an integer greater than 2 such that $(ST)^2 = (ST)^m$, then m is an integer multiple of 8
(D)	If $z \in H$, then $\frac{az+b}{cz+d} \in H$

SECTION 3 (Maximum Marks: 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:
  Full Marks : $+4$ If **ONLY** the correct numerical value is entered in the designated place;
  Zero Marks : 0 In all other cases.

Q.10 Let $\mathbb{N}$ denote the set of all positive integers. Consider the sets

$$A = \{1, 2, 3, 4, 5\} \text{ and } B = \{1, 2, 3, 4, 5, 6, 7\}.$$

Let S be the set of all functions $f: A \to B$ such that $f(2) \neq 2$ and $f(4) \neq 4$. Consider the set

$$T = \{f \in S : \text{there exists a function } g : B \to \mathbb{N} \text{ such that } g(f(x)) = 2^x \text{ for all } x \in A\}.$$

Then the number of elements in the set T is _____.

Q.11 A bookshelf contains 6 distinct books of Mathematics and 5 distinct books of Physics. From these 11 books, 6 books are chosen at random. Let X be the absolute value of the difference between the number of Mathematics books chosen and the number of Physics books chosen. If α is the mean of the random variable X, then the value of 77α is _____.

Q.12 Consider a data consisting of 10 observations $x_1, x_2, \ldots, x_{10}$, whose mean is 5 and variance is 7. If the mean and the variance of the first 8 observations $x_1, x_2, \ldots, x_8$ are 4 and 3.5, respectively, and $x_9 < x_{10}$, then the value of $3x_9 + 2x_{10}$ is _____.

Q.13 Consider the ellipse E given by $\frac{x^2}{18} + \frac{y^2}{12} = 1$. Let H be the hyperbola whose eccentricity is the reciprocal of the eccentricity of E and whose foci are the same as that of E. Let P and Q be the points of intersection of H and the parabola $\sqrt{5}\, y = x^2$ in the first quadrant. Let d be the distance between P and Q.

If a and b are the integers such that $d^2 = a + b\sqrt{5}$, then the value of $a - b$ is _____.

Q.14 For a real number α, let $[\alpha]$ denote the greatest integer less than or equal to α. For a finite set S, let $|S|$ denote the number of elements in the set S.

Consider the functions $f : (-3, 3) \to (-\infty, \infty)$ and $g : (-3, 3) \to (-\infty, \infty)$ defined by

$$f(x) = [x^3] \log_e\left(1 + \sin^2\left(\pi(x - [x])\right)\right)$$

and

$$g(x) = x^3 \sin^2(\pi \log_e(1 + x - [x])).$$

Let

$$A = \{x \in (-3, 3) : f \text{ is discontinuous at } x\}$$

and

$$B = \{x \in (-3, 3) : g \text{ is discontinuous at } x\}.$$

Then the value of $|A| + 2|B| - |A \cap B|$ is _____.

SECTION 4 (Maximum Marks: 8)

- This section contains **TWO (02)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme**:
  Full Marks : +2 If **ONLY** the correct numerical value is entered at the designated place;
  Zero Marks : 0 In all other cases.

Question Stem for Question Nos. 15 and 16

Consider the curve C_1 given by

$$y = e^{-x} \quad \text{for } x \in [0, 10\pi],$$

and the curve C_2 given by

$$y = e^{-x}(\sin x + \cos x) \quad \text{for } x \in [0, 10\pi].$$

Let n be the total number of points of intersection of the curves C_1 and C_2.

Suppose that $\alpha_1, \alpha_2, \ldots, \alpha_n \in [0, 10\pi]$ are the x - coordinates of the points of intersection of the curves C_1 and C_2 such that

$$\alpha_1 < \alpha_2 < \cdots < \alpha_n .$$

Q.15 The value of n is _____.

Q.16 Let β be the area of the region enclosed between the curves C_1, C_2, and the lines $x = \alpha_1$ and $x = \alpha_4$. Then the value of

$$-\frac{1}{\pi}\log_e\left(\beta - 2\,e^{-\frac{\pi}{2}}\right)$$

is _____.

Question Stem for Question Nos. 17 and 18

Consider the ellipses given by

$$x^2 + 4y^2 = 1 \quad \text{and} \quad 4x^2 + y^2 = 1 \ .$$

Q.17 Let P be the point in the first quadrant where the given ellipses intersect. If θ is the acute angle between the tangents to the given ellipses at the point P, then the value of $4\tan\theta$ is _____.

Q.18 If α is the area of the common region that lies inside both the given ellipses, then the value of $\cot\alpha$ is _____.

END OF THE QUESTION PAPER